

Supply and Property Management System for the Supply and Property Management

Office of Camarines Norte State College

A Project Proposal in System Analysis and Design

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ABSTRACT

This study focuses on the design, development, and implementation of the Web-Based Supply and Property Management System for the Supply and Property Management Office (SPMO) of Camarines Norte State College. This project would address the problems brought by manually handling the supply process, such as taking longer hours, lack of precision, and the inability to easily generate government reports. This research used the System Development Life Cycle (SDLC), which includes analysis of systems, modeling, designing, and implementation stages on a UML platform, and is developed on a Laravel PHP platform with a MySQL database. The resulting system has automated purchase requests, automated purchase orders, computerized supply issuance, real-time inventory updates, organized supplier management, and full accountability of properties via ICS and PAR reports. The system has Role-Based Management, generates reports on government-standard documents, is coded with provisions set by Republic Act 9184, streamlines reporting, provides instant updates, includes a user-friendly database with easy-to-read reports, is optimized for multiple platforms, and is user-friendly. The findings from the assessment affirm that the automated system is an efficiency-booster, an error-correcting solution, and a transparency-improving mechanism within the SPMO's different departments. This research concludes that this system is a dependable, scalable, user-friendly solution that brings modernity to the supply and property management practices within Camarines Norte State College. Future improvements may consider integrating with RFID inventory systems, automatic notifications, mobile friendliness, and beefed-up analytics that generate highly precise reports.

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INTRODUCTION

PROJECT BACKGROUND

Higher education institutions rely heavily on supply and inventory management to ensure proper procurement, monitoring, and distribution of office supplies, equipment, and instructional materials. Some of the operations mentioned are primarily handled by the Supply and Property Management Office (SPMO) of the Camarines Norte State College. However, a more methodical and automated approach is required due to the growing volume of supplies handled and the increasing transparency and accountability standards imposed by Philippine laws such as Republic Act (RA) 9184 (Government Procurement Policy Board, 2024).

Conventional paper-based inventory procedures often lead to errors, inefficiencies, and delays. Research shows that inadequate inventory management in educational institutions can result in resource wastage, supply shortages, and even the loss or deterioration of valuable equipment (Madu & Izuchukwu, 2022; Lau, 2007). In contrast, automated and web-based inventory systems that incorporate digital records and real-time tracking significantly improve accuracy, operational efficiency, and regulatory compliance (Ramos-Miller & Pacheco, 2023; Nasution, 2022). Studies in related sectors likewise highlight that automated inventory solutions enhance visibility and reduce discrepancies by providing accurate and timely information to supply officers (Gumilao, 2024).

In addition, RA 9184, the Government Procurement Reform Act, requires public higher education institutions to keep procurement records open, efficient, and electronic, with platforms like PhilGEPS to attain just and accountable procurement processes (PhilGEPS, n.d.). As noted in the literature, compliance with procurement policies, in relation to digital transformation, cements good governance, minimizes

bottlenecks, and paves the way for efficient resource allocation in higher educational institutions (Manuel, 2025; Planergy, 2025; Mncwango & Ramdass, 2023). Thus, CNSC's adoption of a modern supply and property management system increases internal productivity and adheres to national standards in terms of transparency, accountability, and institutional effectiveness.

PURPOSE OF THE STUDY

The Primary purpose of this study is to design and develop a Supply and Property Management System that automates procurement, inventory tracking, and reporting for the Supply and Property Management Office of CNSC.

SCOPE AND LIMITATION

The proposed Supply and Property Management System for the Supply and Property Management Office (SPMO) is designed to automate and streamline the processes of procurement, inventory management, and reporting to support the operations of CNSC. The scope encompasses real-time inventory monitoring and the comprehensive digitization of procurement workflows, including requisition filing, approval hierarchies, and purchase order generation in strict compliance with Republic Act (RA) 9184 and PhilGEPS guidelines.

Additionally, the system will maintain a centralized database for supplier contracts and purchase histories to enhance accountability, while also facilitating advanced reporting and analytics through the generation of the Report of Supplies and Materials Issued (RSMI), Report on the Physical Count of Inventories (RCPI), and consolidated stock card monitoring.

To ensure data integrity and operational efficiency, the solution features automated government-standard PDF generation, role-based security permissions with

transaction audit trails, and a web-based architecture accessible within the CNSC network with the potential for cloud hosting. The system is primarily intended for use by the Supply Officer, Office Assistants, Property Custodians, and Administrative Staff.

While the Limitations are the system's significant improvements are currently bounded by several limitations. A key operational constraint is the lack of full integration with existing accounting systems. Furthermore, external suppliers can only access the system through PhilGEPS. The system's functionality also carries a dependence on specific hardware, such as barcode scanners and a stable internet connection. Implementation also requires comprehensive staff training. Finally, it's noted that advanced features, such as QR CODE or mobile applications, may be added at a later stage rather than being available from the initial launch.

PROJECT DELIVERABLES

The project will deliver a fully functional Web-Based Supply Management System developed using the Laravel framework. This will include an Admin Dashboard for the administrators, while a dedicated user portal will be provided for supply personnel, both equipped with secure authentication and role-based access control. Apart from that, facilities for the automatic generation of necessary supply documents in PDF formats will be provided, including PR or Purchase Requests, PO or Purchase Orders, IAR or Inspection and Acceptance Reports, ICS or Inventory Custodian Slips, PAR or Property Acknowledgement Receipts, RIS or Requisition and Issue Slips, and Stock-Out Reports. Functional modules include Inventory Management for maintaining item and category records, stock-in and stock-out operations, and low-stock monitoring; Procurement Management for supplier records, purchase requests, and purchase orders; and Asset and Property Management for custodianship tracking

and inspection reporting. User Management will handle the different predefined roles and permissions, while Support and Activity Monitoring modules will handle ticketing services, audit logs, and notification services. The technical deliverables of the system are a complete RESTful API for integrating the system, a fully structured MySQL database with all migrations, and all configuration files, routing components, and source code according to the MVC standards of Laravel. These deliverables, when combined, ensure a complete, reliable, and scalable supply management solution.

SYSTEM ANALYSIS

REQUIREMENT ANALYSIS

The Supply and Property Management System (SPMS) is a web-based application designed to digitally transform the inventory, procurement, and property accountability workflow of the CNSC Supply and Property Management Office. By replacing manual, paper-based processes, the SPMS ensures data accuracy, operational transparency, and efficiency in managing government property, supplies, and standardized documentation.

FUNCTIONAL REQUIREMENTS

The system provides the following core capabilities to support the procurement and inventory lifecycle:

Authentication & User Management

Secure Access: Multi-step authentication including secure login, password resets, and account setup via email token links.

Role-Based Access Control (RBAC): Granular permission management (via Spatie) allowing administrators to create, update, and assign roles. This includes mapping legacy is_admin attributes to specific roles.

User Administration: Comprehensive API endpoints and UI for managing user accounts, including listing, creating, updating, and deleting users with associated role metadata.

Procurement Lifecycle (PR & PO)

Purchase Requests (PR): Creation via a multi-step wizard (UI) or API, utilizing unique ID formats (e.g., REQ-YYYY-XXX) with collision handling.

Status tracking (Pending, Approved, Rejected) with automated email notifications upon submission and status changes.

Purchase Orders (PO): Generation of POs from approved PRs, mapping items, suppliers, units of measure (UoM), and payment terms.

Automatic computation of grand totals and capabilities to archive POs by department

Supplier Management: CRUD operations for managing supplier/vendor details linked to POs.

Inventory & Supply Management

Item Management: Categorization and management of supplies with validation for asset types (Non-expendable, Semi-expendable, Expendable).

Stock Operations:

Stock In: Records incoming transactions, increments inventory, updates unit costs, and logs activity.

Stock Out: Decrements inventory based on issuance, enforces minimum thresholds (e.g., preventing issuance if stock $\leq$ 20), and validates availability.

Low-Stock Alerts: Automated reporting based on configurable threshold parameters.

Automated Document Generation

The system automates the generation, preview, and download of government-compliant PDF forms:

Procurement Docs: Purchase Requests (PR) and Purchase Orders (PO).

Accountability Docs: Inventory Custodian Slips (ICS), Inspection and Acceptance Reports (IAR), Property Acknowledgment Receipts (PAR), and Requisition and Issue Slips (RIS).

Flexibility: Supports finding documents by Record ID or generating fallback blank templates.

Reporting & Activity Logging

Audit Trails: comprehensive activity logs for all critical operations (PR submission, Stock In/Out, Item CRUD, login events).

Dashboard & Analytics: Real-time visualization of system metrics and reporting data via API endpoints.

NON - FUNCTIONAL REQUIREMENTS

These requirements ensure the system is secure, reliable, and maintainable:

Security and compliance measures include access control via middleware and RBAC policies, data protection through input validation, server-side sanitation, and login

throttling, and regulatory compliance ensuring output documents meet government standards for audit readiness.

Data consistency and reliability are ensured through transactional integrity for atomic database updates, collision prevention for unique ID generation, and resilient email notifications with TLS fallback and SMTP retry mechanisms.

Performance and usability are achieved through a responsive SPA-based dashboard, efficient server-side processing with optimized APIs, and a refined user experience featuring guided workflows, previews, confirmations, and clear interface feedback.

Maintainability and architecture are supported by a clean MVC structure, an API-first design that enables scalability and future integrations, and the use of standard libraries with maintained test coverage to ensure reliability and ease of extension.

USER REQUIREMENTS

The proposed Supply and Property Management System (SPMS) will serve multiple users in the Supply and Property Management Office (SPMO). Their specific requirements are as follows:

System Administrator: Oversee the entire procurement and inventory lifecycle by enforcing policy and role segregation, maintain system security by managing user access and workflow configurations, and ensure accountability by auditing changes and generating consolidated regulatory reports.

Supply Officer: Must have full access to all modules for monitoring, approval, and oversight. Must generate consolidated reports for higher management and government compliance. Must approve purchase orders and validate.

Office Assistants: provide administrative support by managing supply deliveries, maintaining accurate stock and supplier records, assisting with requisition and purchase orders, and coordinating daily inventory operations.

Student Assistant: Support daily supply operations by assisting with physical receiving, labeling, and moving of stock under strict supervision, perform routine data entry for non-sensitive inventory records, and aid in inventory counts while reporting discrepancies to the designated custodian.

Requester: Initiate departmental procurement needs by creating detailed purchase requests with justifications and specifications, monitor the progress of approvals and deliveries, and ensure accountability by verifying and acknowledging received items.

USER STORIES AND USE CASE DIAGRAM

The Use Case Diagram for Supply and Property Management System illustrates how different users within the Supply and Property Management Office (SPMO) interact with the system. (*You can see Figure 1.*)

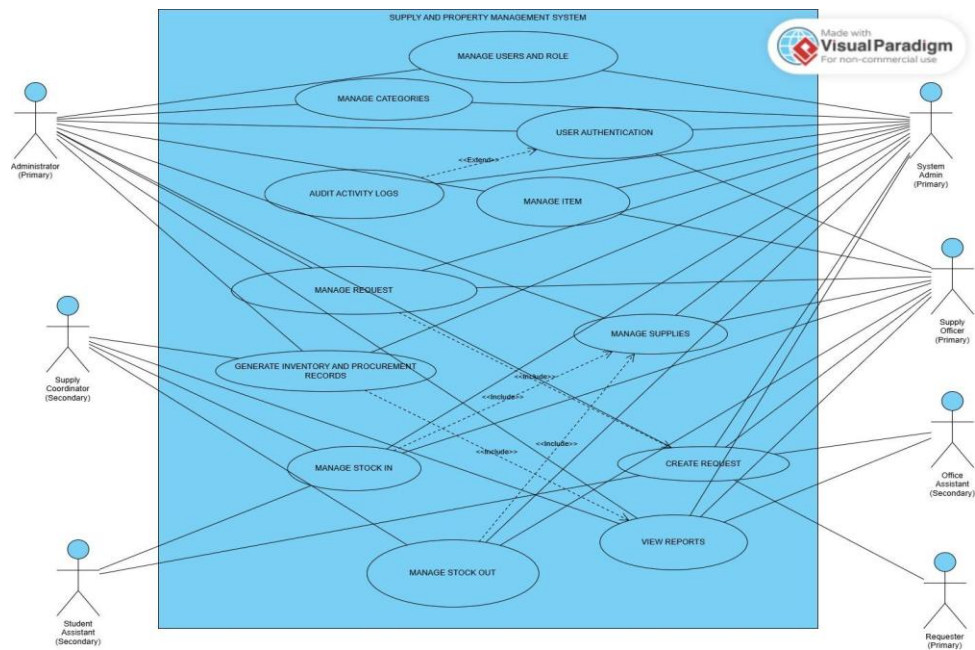


Figure 1: Use Case Diagram of Supply and Property Management System

User Role	User Story	Goal / Purpose
System Admin	As a System Admin, I want to manage everything (categories, items, supplies, stock in/out, requests) so that I can fully control system operations.	To oversee all inventory, procurement, and reporting functions with full access.
Administrator	As an Administrator, I want to manage categories, items, supplies, requests, and view reports so that the system	To ensure proper inventory and procurement

	functions efficiently and remains organized.	management and generate reports for oversight.
Supply Officer	As a Supply Officer, I want to manage items, supplies, create and manage requests, and view reports so that daily operations run smoothly.	To process procurement requests and maintain accurate stock records.
Office Assistant	As an Office Assistant, I want to create requests and view reports so that department supply needs are addressed efficiently.	To assist in the submission of requests and track their status.
Student Assistant	As a Student Assistant, I want to manage stock in/out and create requests so that inventory levels are accurate.	To support stock management and maintain proper records of supplies.
Supply Coordinator	As a Supply Coordinator, I want to manage supplies, stock in/out, create requests, and view reports so that overall supply operations are coordinated.	To ensure smooth inventory flow and maintain accountability in supply management.

Requester	As a Requester, I want to create requests so that my department can receive the needed supplies.	To submit procurement requests and ensure timely fulfillment
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DATA DICTIONARY

User Authentication

This table serves as the central repository for all system actors. It manages authentication (login credentials), authorizes access based on roles (Supply Officer, Staff, etc.), and tracks the status of user accounts.

Column	Type	Attributes	Description
id	BIGINT	PK, Auto Increment	Unique identifier
name	VARCHAR	-	Full name of the user
email	VARCHAR	Unique	Email address
email_verified_at	TIMESTAMP	Nullable	Date email was verified
password	VARCHAR	-	Hashed password
remember_token	VARCHAR	Nullable	"Remember me" token
status	VARCHAR	Default: 'active'	Account status

created_at	TIMESTAMP	-	Creation timestamp
updated_at	TIMESTAMP	-	Last update timestamp

Categories

This table classifies inventory items into logical groups (e.g., "Expendable," "Semi-Expendable"). It is used to organize the stock list and facilitate easier filtering and reporting.

Column	Type	Attributes	Description
id	BIGINT	PK, Auto Increment	Unique identifier
code	VARCHAR (10)	Unique, Nullable	Category code (e.g., C001)
name	VARCHAR	-	Category name
description	TEXT	Nullable	Description of the category
created_at	TIMESTAMP	-	Creation timestamp
updated_at	TIMESTAMP	-	Last update timestamp

Items

This is the master record for all supplies and property. It tracks the current stock quantity, unit costs, and specific details (SKU, description) for every product managed by the SPMO.

Column	Type	Attributes	Description
id	BIGINT	PK, Auto Increment	Unique identifier
sku	VARCHAR	Unique	Stock Keeping Unit
name	VARCHAR	-	Item name
description	TEXT	Nullable	Item description
category_id	BIGINT	FK, Nullable	Links to categories.id
quantity	INTEGER	Default: 0	Current stock quantity
unit	VARCHAR	Nullable	Unit of measure (e.g., pcs, box)
unit_cost	DECIMAL (10,2)	Default: 0.00	Cost per unit
total_value	DECIMAL (15,2)	Default: 0.00	Total value of stock

date	DATE	Nullable	Date of entry/update
created_at	TIMESTAMP	-	Creation timestamp
updated_at	TIMESTAMP	-	Last update timestamp

Supplier

This table maintains a directory of external vendors. It stores contact information and tax details (TIN) required for generating Purchase Orders and evaluating supplier performance.

Column	Type	Attributes	Description
id	BIGINT	PK, Auto Increment	Unique identifier
name	VARCHAR	-	Supplier name
address	TEXT	Nullable	Supplier address
tin	VARCHAR	Nullable	Tax Identification Number
contact	VARCHAR	Nullable	Contact number

email	VARCHAR	Nullable	Email address
latitude	DECIMAL (10,8)	Nullable	Geo-location latitude
longitude	DECIMAL (11,8)	Nullable	Geo-location longitude
created_at	TIMESTAMP	-	Creation timestamp
updated_at	TIMESTAMP	-	Last update timestamp
deleted_at	TIMESTAMP	Nullable	Soft delete timestamp

Stock In

This table logs the history of all deliveries. It records when inventory was added to the system, from which supplier, and at what cost, ensuring an audit trail for inventory increases.

Column	Type	Attributes	Description
id	BIGINT	PK, Auto Increment	Unique identifier

transaction_id	VARCHAR	Unique	Transaction reference ID
sku	VARCHAR	-	SKU of item received
product_name	VARCHAR	-	Name of item received
quantity	INTEGER	-	Quantity received
unit_cost	DECIMAL (10,2)	Default: 0.00	Cost at time of receipt
supplier	VARCHAR	Nullable	Supplier name
received_by	VARCHAR	Nullable	Name of receiver
date_received	DATE	-	Date stock was received
created_at	TIMESTAMP	-	Creation timestamp
updated_at	TIMESTAMP	-	Last update timestamp

Stock out

This table logs the history of all issuances. It records when inventory was removed from the warehouse, who it was issued to, and for what purpose, ensuring an audit trail for inventory decreases.

Column	Type	Attributes	Description
id	BIGINT	PK, Auto Increment	Unique identifier
transaction_id	VARCHAR	Nullable	Transaction reference ID
issue_id	VARCHAR	Unique, Nullable	Issue Slip ID
sku	VARCHAR	-	SKU of item issued
product_name	VARCHAR	-	Name of item issued
quantity	INTEGER	-	Quantity issued
unit_cost	DECIMAL (10,2)	Default: 0.00	Cost per unit
total_cost	DECIMAL (12,2)	Default: 0.00	Total cost of issued items
department	VARCHAR	Nullable	Department issued to

issued_to	VARCHAR	Nullable	Person issued to
issued_by	VARCHAR	Nullable	Person who issued
purpose	TEXT	Nullable	Purpose of issuance
date_issued	DATE	-	Date of issuance
created_at	TIMESTAMP	-	Creation timestamp
updated_at	TIMESTAMP	-	Last update timestamp

Purchase Request

This table stores internal requests from various departments for the acquisition of goods. It tracks the approval workflow before a request is converted into an official Purchase Order.

Column	Type	Attributes	Description
id	BIGINT	PK, Auto Increment	Unique identifier
request_id	VARCHAR	Unique	Request Reference ID

requester	VARCHAR	Nullable	Name of requester
designation	VARCHAR	Nullable	Job title/Designation
department	VARCHAR	Nullable	Department
items	JSON	Nullable	List of items requested
purpose	TEXT	Nullable	Purpose of purchase
unit	VARCHAR	Nullable	Unit of measure
quantity	INTEGER	Nullable	Quantity requested
unit_cost	DECIMAL (12,2)	Nullable	Estimated unit cost
total_cost	DECIMAL (12,2)	Nullable	Estimated total cost
needed_date	DATE	Nullable	Date items are needed
priority	VARCHAR	Default: 'Low'	Priority level
status	VARCHAR	Default: 'Incoming'	Request status

submitted_at	TIMESTAMP	Nullable	Submission timestamp
metadata	JSON	Nullable	Additional data
created_at	TIMESTAMP	-	Creation timestamp
updated_at	TIMESTAMP	-	Last update timestamp

Purchase Orders

This table stores the official contracts between the agency and the suppliers. It contains all legal details required by RA 9184, including delivery terms, payment terms, and funding validation.

Column	Type	Attributes	Description
id	BIGINT	PK, Auto Increment	Unique identifier
po_number	VARCHAR	Unique	PO Number
supplier	VARCHAR	Nullable	Supplier Name
supplier_address	TEXT	Nullable	Supplier Address
date_of_purchase	DATE	Nullable	Purchase Date

tin_number	VARCHAR	Nullable	TIN
mode_of_procurement	VARCHAR	Nullable	Mode (e.g., Bidding, Shopping)
place_of_delivery	VARCHAR	Nullable	Delivery location
delivery_term	VARCHAR	Nullable	Delivery terms
date_of_delivery	VARCHAR	Nullable	Delivery date/period
payment_term	VARCHAR	Nullable	Payment terms
items	JSON	Nullable	List of items ordered
grand_total	DECIMAL (15,2)	Default: 0.00	Total PO amount
fund_cluster	VARCHAR	Nullable	Fund Cluster
ors_burs_no	VARCHAR	Nullable	ORS/BURS Number
funds_available	VARCHAR	Nullable	Funds Available certifier
ors_burs_date	DATE	Nullable	ORS/BURS Date
ors_burs_amount	DECIMAL (15,2)	Nullable	Amount obligated

accountant_signature	VARCHAR	Nullable	Path to signature
entity_name	VARCHAR	Nullable	Agency/Entity Name
entity_address	TEXT	Nullable	Agency Address
status	VARCHAR	Default: 'Draft'	PO Status
department	VARCHAR	Nullable	Department
gentlemen	TEXT	Nullable	Salutation/Header text
notes	TEXT	Nullable	Additional notes
created_at	TIMESTAMP	-	Creation timestamp
updated_at	TIMESTAMP	-	Last update timestamp

Requisition Issue Slip (RIS)

This table manages the documentation for the issuance of supplies from the warehouse. It serves as proof that the end-user has successfully received the items they requested.

Column	Type	Attributes	Description
id	BIGINT	PK, Auto Increment	Unique identifier
purchase_order_id	BIGINT	FK, Nullable	Links to purchase_orders.id
ris_no	VARCHAR	Unique	RIS Number
entity_name	VARCHAR	Nullable	Entity Name
fund_cluster	VARCHAR	Nullable	Fund Cluster
division	VARCHAR	Nullable	Division
responsibility_center_code	VARCHAR	Nullable	RCC
office	VARCHAR	Nullable	Office
purpose	TEXT	Nullable	Purpose
items	JSON	Nullable	Items requested/issued
requested_by_*	VARIOUS	-	Name, Signature, Designation, Date
approved_by_*	VARIOUS	-	Name, Signature, Designation, Date

issued_by_*	VARIOUS	-	Name, Signature, Designation, Date
received_by_*	VARIOUS	-	Name, Signature, Designation, Date
status	VARCHAR	Default: 'Draft'	Status
created_at	TIMESTAMP	-	Creation timestamp
updated_at	TIMESTAMP	-	Last update timestamp
id	BIGINT	PK, Auto Increment	Unique identifier
purchase_order_id	BIGINT	FK, Nullable	Links to purchase_orders.id
ris_no	VARCHAR	Unique	RIS Number
entity_name	VARCHAR	Nullable	Entity Name
fund_cluster	VARCHAR	Nullable	Fund Cluster
division	VARCHAR	Nullable	Division
responsibility_center_code	VARCHAR	Nullable	RCC

office	VARCHAR	Nullable	Office
purpose	TEXT	Nullable	Purpose
items	JSON	Nullable	Items requested/issued
requested_by_*	VARIOUS	-	Name, Signature, Designation, Date
approved_by_*	VARIOUS	-	Name, Signature, Designation, Date
issued_by_*	VARIOUS	-	Name, Signature, Designation, Date
received_by_*	VARIOUS	-	Name, Signature, Designation, Date
status	VARCHAR	Default: 'Draft'	Status
created_at	TIMESTAMP	-	Creation timestamp
updated_at	TIMESTAMP	-	Last update timestamp

Inspection Acceptance Report (IAR)

This table records the inspection results of delivered goods. It validates that items meet the specifications and quantity ordered before they are officially accepted into inventory.

Column	Type	Attributes	Description
id	BIGINT	PK, Auto Increment	Unique identifier
purchase_order_id	BIGINT	FK, Nullable	Links to purchase_orders.id
iar_no	VARCHAR	Unique	IAR Number
entity_name	VARCHAR	Nullable	Entity Name
fund_cluster	VARCHAR	Nullable	Fund Cluster
supplier	VARCHAR	Nullable	Supplier
iar_date	DATE	Nullable	Report Date
po_no	VARCHAR	Nullable	PO Number Reference
po_date	DATE	Nullable	PO Date Reference

requisitioning_office	VARCHAR	Nullable	Office
responsibility_center_code	VARCHAR	Nullable	RCC
invoice_no	VARCHAR	Nullable	Invoice Number
invoice_date	DATE	Nullable	Invoice Date
date_inspected	DATE	Nullable	Inspection Date
date_received	DATE	Nullable	Receipt Date
inspection_status	VARCHAR	Nullable	Status of inspection
acceptance_status	VARCHAR	Nullable	Status of acceptance
items	JSON	Nullable	Items inspected
status	VARCHAR	Default: 'Draft'	Status
created_at	TIMESTAMP	-	Creation timestamp
updated_at	TIMESTAMP	-	Last update timestamp

Inventory Custodian Slip (ICS)

This table tracks the issuance of "Semi-Expendable Property" (items with a useful life of more than one year but below the capitalization threshold) to specific custodians for accountability.

Column	Type	Attributes	Description
id	BIGINT	PK, Auto Increment	Unique identifier
purchase_order_id	BIGINT	FK, Nullable	Links to purchase_orders.id
ics_no	VARCHAR	Unique	ICS Number
entity_name	VARCHAR	Nullable	Entity Name
fund_cluster	VARCHAR	Nullable	Fund Cluster
items	JSON	Nullable	Items issued
grand_total	DECIMAL (15,2)	Default: 0.00	Total Value
status	VARCHAR	Default: 'Draft'	Status
received_from_*	VARIOUS	-	Name, Position, Date
received_by_*	VARIOUS	-	Name, Position, Date
created_at	TIMESTAMP	-	Creation timestamp

updated_at	TIMESTAMP	-	Last update timestamp
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Property Acknowledge Receipt (PAR)

This table tracks the issuance of high-value Property, Plant, and Equipment (PPE) to employees. It serves as the legal record of accountability for government assets.

Column	Type	Attributes	Description
id	BIGINT	PK, Auto Increment	Unique identifier
purchase_order_id	BIGINT	FK, Nullable	Links to purchase_orders.id
par_no	VARCHAR	Unique	PAR Number
entity_name	VARCHAR	Nullable	Entity Name
fund_cluster	VARCHAR	Nullable	Fund Cluster
date	DATE	Nullable	Date
items	JSON	Nullable	Items issued
grand_total	DECIMAL (15,2)	Nullable	Total Value

received_by_*	VARIOUS	-	Name, Position, Date
issued_by_*	VARIOUS	-	Name, Position, Date
status	VARCHAR	Default: 'Draft'	Status
created_at	TIMESTAMP	-	Creation timestamp
updated_at	TIMESTAMP	-	Last update timestamp

Support Tickets

This table manages user-reported issues and inquiries. It tracks the status of help requests to ensure timely resolution of system problems.

Column	Type	Attributes	Description
id	BIGINT	PK, Auto Increment	Unique identifier
support_ticket_id	BIGINT	FK	Links to support_tickets.id
filename	VARCHAR	-	Stored filename
original_name	VARCHAR	-	Original filename

mime	VARCHAR	Nullable	MIME type
size	BIGINT	Default: 0	File size in bytes
created_at	TIMESTAMP	-	Creation timestamp

Support Attachments

This table stores file uploads (such as screenshots or error logs) related to support tickets, helping administrators diagnose technical issues.

Column	Type	Attributes	Description
id	BIGINT	PK, Auto Increment	Unique identifier
support_ticket_id	BIGINT	FK	Links to support_tickets.id
filename	VARCHAR	-	Stored filename
original_name	VARCHAR	-	Original filename
mime	VARCHAR	Nullable	MIME type
size	BIGINT	Default: 0	File size in bytes
created_at	TIMESTAMP	-	Creation timestamp

Site Contents

This table stores dynamic configuration settings and content (e.g., system logos, header text, contact info) allowing admins to update site details without modifying the source code.

Column	Type	Attributes	Description
id	BIGINT	PK, Auto Increment	Unique identifier
key	VARCHAR	Unique	Config key
value	JSON	Nullable	Config value
created_at	TIMESTAMP	-	Creation timestamp
updated_at	TIMESTAMP	-	Last update timestamp

Activity Log

This table serves as the system's audit trail. It automatically records all user actions (creations, updates, deletions) to ensure security, accountability, and traceability of transactions.

Column	Type	Attributes	Description
id	BIGINT	PK, Auto Increment	Unique identifier
log_name	VARCHAR	Default: 'default'	Log category
description	TEXT	-	Activity description
subject_type	VARCHAR	Nullable	Model class name
subject_id	VARCHAR	Nullable	Model ID
causer_type	VARCHAR	Nullable	User class name
causer_id	VARCHAR	Nullable	User ID
properties	JSON	Nullable	Extra data
batch_uuid	VARCHAR	Nullable	Batch ID
created_at	TIMESTAMP	-	Creation timestamp
updated_at	TIMESTAMP	-	Last update timestamp

SYSTEM DESIGN

SYSTEM ARCHITECTURE

The Supply and Property Management System is designed as a robust, web-based solution designed to modernize the procurement, inventory, and asset tracking workflows of the organization. The system is built to address the limitations of manual record-keeping by enforcing digital accountability, ensuring real-time data accuracy, and automating the generation of government-mandated forms.

ARCHITECTURE DIAGRAM

The system implements the Model-View-Controller (MVC) architectural pattern. This design choice ensures a clean separation of concerns, which is critical for the maintainability and scalability of the application. (*You can see Figure 2.*)

Model (Data Layer): Manages business logic and database interactions. It represents real-world entities such as `PurchaseRequest`, `Item`, and `InventoryCustodianSlip`. This layer allows the system to enforce data integrity rules directly at the object level.

View (Presentation Layer): Responsible for the user interface. Utilizing **Laravel Blade** templates combined with **Tailwind CSS**, the view layer presents data in a responsive format, ensuring accessibility across different devices.

Controller (Application Logic): Acts as the intermediary, handling user input, invoking the appropriate Model logic, and determining the correct View to render. This separation allows for modular development where backend logic can be updated without breaking the frontend interface.

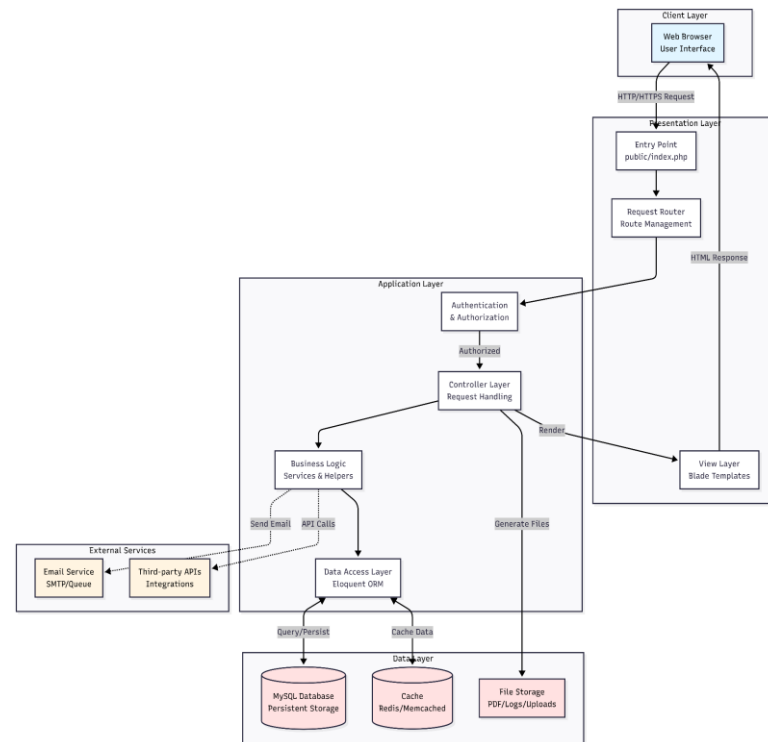


Figure 2: System Architecture Diagram of Supply and Property Management System

DATABASE DESIGN (ERD)

The Supply and Property Management System acts as a secure platform centered around the **Users** entity. This table serves as the primary authentication source, storing user credentials and profile details. It maintains a one-to-many relationship with the **Activity Log**, ensuring that every critical action performed by a user is recorded for security and auditing purposes. Additionally, the system includes a support framework where users can raise issues through **Support Tickets**. This entity is further linked to **Support Attachments**, allowing users to upload multiple files or evidence for a single ticket, creating a clear hierarchy of User → Ticket → Attachments. (You can see on Figure 3.)

The inventory section is designed to track the lifecycle of physical assets. The core entity is **Items**, which serves as the master record for all products, holding static data like SKUs and descriptions. To organize these items, the system uses a **Categories**

table (e.g., Office Supplies, Hardware) linked via a one-to-many relationship. Instead of constantly modifying the item record for quantity changes, the system uses a transactional approach with **Stock in** and **Stock Out** tables. These entities track the history of inventory movement, recording exactly when items were received from suppliers or issued to departments, which allows for accurate historical reporting and stock level calculation. (You can see on Figure 3.)

This module digitizes the standard government or corporate procurement process. It begins with **Purchase Requests**, where departments identify their needs. These requests are processed into **Purchase Orders (PO)**, which acts as the central hub for financial tracking. Once goods are delivered, the PO links to **Inspection Acceptance Reports**, confirming the quality and quantity of the delivery. Finally, the system manages asset accountability through three specific issuance documents linked to the PO: **Requisition Issue Slips (RIS)** for consumable items, **Inventory Custodian Slips (ICS)** for semi-expendable items, and **Property Acknowledgement Receipts (PAR)** for high-value assets. This ensures a complete paper trail from the initial request to the final assignment of responsibility to an employee. (You can see on Figure 3.)

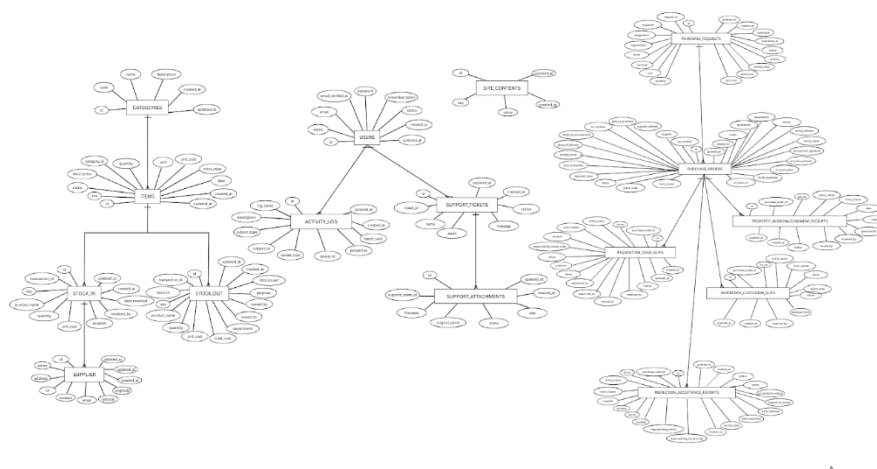


Figure 3: Entity Relationship Diagram of Supply and Property Management System

DATA FLOW DIAGRAM

The DFD Level 0, or Context Diagram, illustrates the "Big Picture" by treating the entire Supply and Property Management System as a single "black box" (Process 0.0) that interacts with four key external entities. The process initiates with the Employee/User, who submits Purchase Requests or Requisition Issue Slips into the system and subsequently receives the requested items along with necessary accountability forms such as the PAR or ICS. Central to this operation is the Admin/Supply Officer, who serves as the main operator by feeding the system with inventory data and user management inputs while receiving generated reports, such as the IAR, PO, and inventory summaries, as well as status notifications in return. The workflow also involves the Approving Authority acting as a gatekeeper; the system sends them pending requests, and they return an "Approved" status to allow the process to continue. Finally, the Supplier interacts at the procurement stage by receiving Purchase Orders from the system and responding with the physical delivery and invoice, which are then recorded by the system. (*You can see on **Figure 4.***)

The DFD Level 1 decomposes the system into five major internal functions, beginning with Process 1.0, Manage User Access, where credentials provided by users and admins are verified against the Users Data Store (D1) to grant role-specific feature access. The workflow transitions to procurement in Process 2.0, where a User submits a Purchase Request stored in the PR Data Store (D2); once cleared by the Approving Authority, the Admin converts this into a Purchase Order, which is saved in the PO Data Store (D3) and transmitted to the Supplier. Upon the delivery of goods, Process 3.0, Inspection & Acceptance, engages as the admin validates the items and generates an Inspection Acceptance Report saved in the IAR Data Store (D5), an action that automatically updates the Inventory Data Store (D4) to reflect the new stock.

Simultaneously, Process 4.0, Manage Inventory, allows the admin to maintain data integrity by adding or updating item details directly within the Inventory Data Store (D4), ensuring it remains the central source of truth. Finally, Process 5.0, Manage Issuance, handles distribution when a User submits a Requisition Issue Slip; the system checks availability in the Inventory Data Store (D4) and, upon approval, generates the required PAR or ICS accountability forms which are saved to Issuance Records (D6), effectively deducting the items from inventory and finalizing the physical release to the user. (You can see on Figure 4.1)

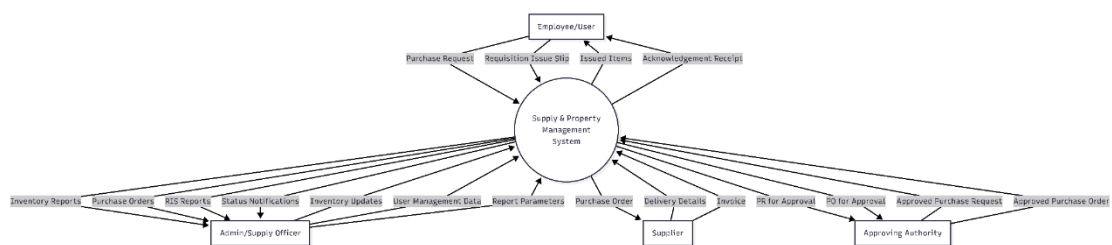


Figure 4: Data flow Diagram Level 0 of Supply and Property Management System

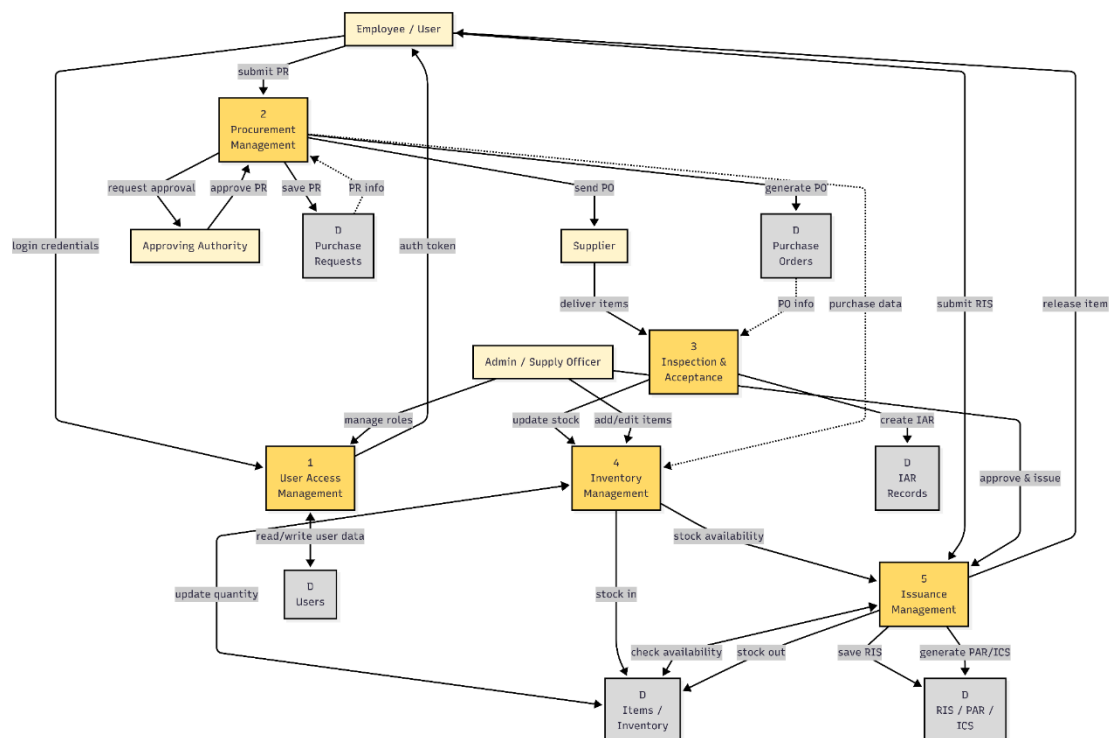


Figure 5: Data flow Diagram Level 1 of Supply and Property Management System

USER INTERFACE DESIGN

The Supply and Property Management System (SPMS) feature a cohesive, modular interface design built to support high-volume data entry and real-time monitoring. The design language prioritizes clarity through semantic coloring, progressive disclosure (wizards and modals), and interactive feedback. (*See Appendices A.*)

IMPLEMENTATION AND TESTING

TECHNOLOGY STACK

The Supply System project was developed using a modern and efficient full-stack web development environment centered on Laravel and Tailwind CSS. The stack was selected to ensure fast performance, scalability, modularity, and ease of maintenance. Based on the analysis of the project files, including *composer.json*, *package.json*, and the directory structure.

Category	Technology / Tool	Justification
Backend Language	PHP 8.2+	Required for Laravel; supports modern features.
Backend Framework	Laravel 12	Provides MVC, ORM, routing, security, and tooling.
Testing Framework	Pest	Modern, expressive testing for Laravel apps.

Frontend Template	Blade	Simplifies dynamic HTML rendering.
Styling	Tailwind CSS 4	Utility-first styling for rapid UI development.
Build Tool	Vite	Fast asset bundling and HMR.
Frontend Scripting	JavaScript (ESM)	Adds interactivity to the UI.
Database	MySQL / MariaDB	Relational data management with Laravel support.
RBAC	Spatie Permission	Role and permission management.
Audit Logging	Spatie Activitylog	Logs user activities and changes for accountability.
Reports (PDF)	DOMPDF, jsPDF, html2canvas	Backend and frontend PDF generation.
Reports (Excel)	exceljs	Exports detailed spreadsheets.
Charts	FusionCharts	Interactive data visualization.

HTTP Client	Axios	Asynchronous calls to the backend API.
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SYSTEM MODULES

The Supply and Property Management System (SPMS) is composed of multiple interconnected modules, each designed to perform specific functions that collectively automate procurement, inventory management, document generation, and reporting. The modules are structured to support role-based access, maintain audit trails, and ensure regulatory compliance with government standards.

Module	Core Logic	Functions / Features
Authentication / Access System	Serves as the gateway and onboarding system.	This module allows users to log in and log out through the AccessController, set up their accounts via the AccountSetupController, and securely reset passwords using the PasswordResetController. The frontend is managed using access-system.blade.php.

Authorization / Roles & Permissions	Implements role-based access control (RBAC).	This module manages user roles and permissions, initializes them using PermissionSeeder.php, enforces access rules via model policies, and synchronizes roles and permissions during deployment with SyncRolesPermissions.php.
Users & Admin Dashboard	Administrative hub for user management and system monitoring.	This module allows administrators to manage users and their roles, view high-level system metrics such as pending purchase requests and low-stock alerts, and visualize data dynamically using dashboard.js.
Inventory Management (Items, Categories, Stock)	Manages all physical goods and stock movements.	This module categorizes items for easier reporting, links items with their suppliers, tracks stock-in and stock-out transactions to ensure auditability and provides API endpoints for asynchronous frontend requests.
Procurement Workflow (PR & PO)	Implements a state-machine workflow for procurement.	This module manages purchase requests through stages from Draft to Completed, triggers automated actions via PurchaseRequestObserver.php

		when statuses change, converts approved purchase requests into purchase orders using PurchaseOrderController, and sends email notifications for pending tasks.
Document Generation (PDFs)	Produces government-compliant official documents.	This module generates PDF documents using barryvdh/laravel-dompdf, including key forms such as the IAR (Inspection and Acceptance Report) and RIS (Requisition and Issue Slip), with Blade views styled to replicate official government forms.
Activity / Audit Logging	Tracks user actions for accountability.	This module records all create, update, and delete actions using Spatie/Activitylog, allows custom logging of non-model events through activity_helper.php, and provides an administrative view for reviewing user activities.
Support & Site Content	Handles internal support and dynamic static pages.	This module enables users to submit support tickets using SupportTicket and SupportAttachment, while allowing administrators to edit website content

		such as “About Us” or “Terms of Service” without developer intervention.
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TEST PLAN AND TEST CASE

The Test Plan and Test Cases provide a structured approach to verify that the Web-Based Supply and Property Management System functions accurately, reliably, and efficiently. Testing ensures critical modules, such as Inventory Management, Product and Category Management, and Stock In/Out operations, perform correctly under normal and concurrent usage. Scenarios include adding, updating, and deleting items, managing stock, and handling errors. The system was tested in a Laravel-MySQL environment with a responsive frontend across multiple browsers to ensure compatibility. Functional, validation, integration, and parallel testing techniques were applied, with clear steps and expected results for each test case, ensuring data integrity, system reliability, and readiness for deployment.

Test Case 1: Authentication

This set of test cases verifies the authentication functionality of the system, including successful and failed logins, rate-limiting for excessive login attempts, and proper logout behavior. It ensures secure access and session management for users.

Test Case ID	Feature	Test Scenario	Pre-conditions	Test Step	Expected Result	Type
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AUTH-001	Authentication	Login Success	1. User exists.	1. POST /login with valid credentials.	Response 200 / redirect; user authenticated.	Happypath
AUTH-002	Authentication	Login Failure	1. User exists.	1. POST /login with invalid credentials.	Response 422/validation error or error message.	Happypath
AUTH-003	Authentication	Rate Limit Block	1. User exists.	1. Attempt >5 login attempts within 1 minute.	Requests throttled; 429 response or rate limit message.	Happypath
AUTH-004	Authentication	Logout	1. User is authenticated.	1. POST /logout after login.	Session invalidated; redirect to	Happypath

					login page or 200.	
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Test Case 2: Account Setup

These test cases verify the account setup process, ensuring that invited users can successfully activate their account with a valid token and that invalid or expired tokens are properly rejected.

Test Case ID	Feature	Test Scenario	Pre-conditions	Test Step	Expected Result	Type
ACC-001	Account Setup	Accept Account Setup	1. Unconfigured invited user with valid token.	1. Visit <code>account/setup/{token}</code> . 2. Submit form. 3. Log in.	Account activated; can login.	Happy path
ACC-002	Account Setup	Invalid Token	1. Invalid or expired token.	1. Visit <code>/account/setup/{invalid_token}</code> .	Error message; form rejected; no	unHappy path

				2. Submit form.	activation .	
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Test Case 3: Password Reset

These test cases verify the password reset functionality, including requesting a reset, performing the reset with a valid token, and handling expired or invalid tokens securely.

Test Case ID	Feature	Test Scenario	Pre-conditions	Test Step	Expected Result	Type
PW-001	Password Reset	Request Reset	1. User exists.	1. POST /forgot-password. 2. Receive token via email.	Reset token sent to email; 200 response.	Happy path

PW-002	Password Reset	Perform Reset	1. Valid reset token.	1. POST /reset-password with token & new password. 2. Attempt login.	Password updated; login succeeds.	Happypath
PW-003	Password Reset	Expired/Invalid Token	1. Expired/invalid reset token.	1. POST /reset-password with expired/invalid token.	400/422 response; password unchanged; error message.	unHappypath

Test Case 4: Users API

These test cases verify the Users API for administrative operations, including creating users, updating their roles and permissions, and deleting users, ensuring proper access control and system integrity.

Test Case ID	Feature	Test Scenario	Pre-conditions	Test Step	Expected Result	Type
USE R-001	Users API	Create User	1. Admin account. 2. Valid payload.	1. POST /api/users with payload.	201 created; user returned.	Happypath
USE R-002	Users API	Update User & Role	1. Admin account. 2. User exists.	1. PATCH /api/users/{id}. 2. Update role & perms.	200; roles updated and permissions enforced.	Happypath
USE R-003	Users API	Delete User	1. Admin account. 2. User exists.	1. DELETE /api/users/{id}.	204; user removed and cannot login.	Happypath

Test Case 5: Roles

These test cases validate role management and permission enforcement, ensuring that roles can be created and assigned correctly and that unauthorized users are prevented from accessing restricted routes.

Test Case ID	Feature	Test Scenario	Pre-conditions	Test Step	Expected Result	Type
ROL E-001	Roles	Create & Assign Role	1. Admin account. 2. Permissions exist.	1. Create role. 2. Assign permission. 3. Attempt access.	Role/permission applied; unauthorized blocked.	Happypath
ROL E-002	Roles	Unauthorized Access	1. Non-admin user.	1. Attempt to access admin-only route.	403 Forbidden or redirect;	Happypath

			2. No permission.		access denied.	
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Test Case 6: Categories

These test cases verify the Categories API, ensuring that admins can perform CRUD operations successfully and that invalid data is properly validated and rejected.

Test Case ID	Feature	Test Scenario	Pre-conditions	Test Step	Expected Result	Type
CAT-001	Categories	CRUD Category	1. Admin token. 2. Valid payload.	1. POST / GET / PATCH / DELETE .	CRUD success; valid responses.	Happy path

CAT-002	Categories	Validation Failure	1. Admin token. 2. Invalid payload.	1. POST /api/categories with invalid data.	422 validation errors; create prevented.	unHappy path
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Test Case 7: Items

These test cases validate the Items API by ensuring that CRUD operations work correctly, low-stock items are properly listed, and duplicate SKUs are rejected to maintain data integrity.

Test Case ID	Feature	Test Scenario	Pre-conditions	Test Step	Expected Result	Type
ITEM-001	Items	CRUD Items	1. Admin token. 2. Valid item payload.	1. POST / GET / PATCH / DELETE.	CRUD success; low-stock endpoint	Happy path

				2. Check low stock.	lists items.	
ITEM-002	Items	Duplicate SKU Rejected	1. Admin token. 2. Existing SKU.	1. POST /api/items with existing SKU.	422 validation error; item not created.	unHappy path

Test Case 8: Supplier

These test cases validate the Suppliers API, ensuring that CRUD operations function correctly and that invalid supplier data is properly rejected to maintain data integrity.

Test Case ID	Feature	Test Scenario	Pre-conditions	Test Step	Expected Result	Type
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SUP-001	Suppliers	Supplier CRUD	1. Admin token. 2. Valid payload.	1. POST / GET / PATCH / DELETE.	CRUD success; validation s enforced.	Happy path
SUP-002	Suppliers	Supplier Validation Failure	1. Admin token. 2. Invalid payload.	1. POST /api/suppliers with invalid data.	422 validation errors; create prevented.	unHappy path

Test Case 9: Activity Logs

These test cases verify the Activity Logs feature, ensuring that activities are recorded correctly, can be filtered, and include essential details like actor, timestamp, and subject.

Test Case ID	Feature	Test Scenario	Pre-conditions	Test Step	Expected Result	Type
ACT-001	Activity Logs	Create & Filter Activities	1. Activity model present.	1. POST activity. 2. GET with filters.	Activity stored; filters return results.	Happypath
ACT-002	Activity Logs	Actor & Timestamp Present	1. Activity exists.	1. POST activity. 2. GET activity by ID.	Activity contains actor, timestamp, and subject.	Happypath

Test Case 10: Support

These test cases validate the Support feature, ensuring users can create tickets with attachments, update statuses, and view tickets along with associated comments.

Test Case ID	Feature	Test Scenario	Pre-conditions	Test Step	Expected Result	Type
SUPPORT-001	Support	Create Ticket with Attachment	1. Authenticated user. 2. Attachments allowed.	1. POST support ticket. 2. POST attachment. 3. Update status.	Ticket created, attachment saved, status updated.	Happy path

SUPPORT-002	Support	View Ticket & Comments	1. Authenticated user. 2. Existing ticket.	1. GET /support-tickets/{id}. 2. GET comments.	200; ticket details and comments returned.	Happy path
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Test Case 11: Site Content

These test cases validate site content management, ensuring that admins can update content successfully while unauthorized users are prevented from making changes.

Test Case ID	Feature	Test Scenario	Pre-conditions	Test Step	Expected Result	Type
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SIT E- 001	Site Content	Get & Update Site Content	1. Admin token.	1. GET /api/site- contents/{ key}. 2. PUT updated content.	Content updated & returned.	Happyp ath
SIT E- 002	Site Content	Unauthor ized Update Rejected	1. Non- admin user.	1. PUT /api/site- contents/{ key} as non- admin.	403 Forbidden; content unchange d.	Happyp ath

Test Case 11: Purchase Request

These test cases verify the Purchase Request feature, covering API and UI creation, field validations, PDF generation, status notifications, and updating PR items to ensure proper functionality and data integrity.

Test Case ID	Feature	Test Scenario	Pre-conditions	Test Step	Expected Result	Type
PR-001	Purchase Request	Create PR (API)	1. API token. 2. Items exist.	1. POST /api/purchase-requests with items.	201 created; PR stored with items.	Happy path
PR-002	Purchase Request	Create PR (UI)	1. Authenticate requester. 2. Items exist.	1. Navigate UI. 2. Fill & submit form.	PR created and appears in user list.	Happy path
PR-003	Purchase Request	Validate PR Required Fields	1. Authenticate user.	1. Submit with missing fields.	Validation messages; 422 response.	Negative
PR-004	Purchase Request	Generate PR PDF	1. PR exists.	1. POST generate or	PDF content correct.	Happy path

				GET preview.		
PR-005	Purchase Request	PR Status change notifications	1. PR exists. 2. Approvers exist.	1. Update status. 2. Assert notifications.	Mail queued; notifications recorded.	Integration
PR-006	Purchase Request	Update PR & Items	1. PR exists. 2. Requester.	1. PATCH /purchase-requests/{id} to change items/quantity.	200; PR updated; totals recalculated.	Happy path

Test Case 12: Purchase Order

These test cases verify the Purchase Request feature, covering API and UI creation, field validations, PDF generation, status notifications, and updating PR items to ensure proper functionality and data integrity.

Test Case ID	Feature	Test Scenario	Pre-conditions	Test Step	Expected Result	Type
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PO-001	Purchase Order	Create PO (API)	1. API token for PO user. 2. Items exist.	1. POST /api/purchase-orders.	201 created; PO returned with items/totals.	Happypath
PO-002	Purchase Order	Create PO from PR (UI)	1. Approved PR exists.	1. From PR, create PO. 2. Add supplier/costs.	PO created; items carried.	Happypath
PO-003	Purchase Order	Update PO to Received & Inventory	1. PO with items.	1. Update PO status to received. 2. Validate inventory & IAR.	Inventory updated; IAR created.	Happypath
PO-004	Purchase Order	Generate PO PDF	1. PO exists.	1. GET /purchase-	PDF contains PO details.	Happypath

				order/{id}/pdf		
				.		
PO-005	Purchase Order	Cancel PO Prevents Inventory Updates	1. PO exists; not received.	1. POST /purchase-orders/{id}/status with cancelled.	200; PO cancelled; inventory not affected.	Happy path

Test Case 12: IAR

These test cases verify the Inventory Accomplishment Report (IAR) functionality, ensuring IARs can be generated from POs, custodian details can be updated, and PDFs correctly reflect the data.

Test Case ID	Feature	Test Scenario	Pre-conditions	Test Step	Expected Result	Type
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IAR -001	IAR	Generate IAR for PO	1. Received PO. 2. Items present.	1. Create IAR from PO. 2. Fill custodi an. 3. Genera te PDF.	IAR created and PDF valid.	Happypa th
IAR -002	IAR	Edit IAR Updates Custodia n	1. IAR exists.	1. PATC H /invent ory- custodi an- slip/{id } change	200; custodian updated; PDF reflects new custodian.	Happypa th

				custodi an. 2. Genera te PDF.		
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Test Case 13: PAR

These test cases verify the Property Acknowledgement Receipt (PAR) functionality, ensuring PARs can be created, PDFs generated, and custodians reassigned while updating asset assignments.

Test Case ID	Feature	Test Scenario	Pre-conditions	Test Step	Expected Result	Type
PAR-001	PAR	Create PAR & Generate PDF	1. Items received/assigned.	1. Create PAR. 2. Generate PDF.	PAR created; PDF valid.	Happypath

PAR-002	PAR	Reassign/Change Custodian	1. PAR exists.	1. PATCH /property-acknowledgement-receipt/{id} change user.	200; PAR updated; asset assignment updated.	Happypath
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Test Case 14: Inspection

This test case verifies the inspection process, ensuring that an IAR/Inspection Acceptance PDF can be generated correctly with all inspection results and related items included.

Test Case ID	Feature	Test Scenario	Pre-conditions	Test Step	Expected Result	Type
IAR-002	Inspection	Generate IAR/Inspection Acceptance PDF	1. Received PO with inspection .	1. Create inspection. 2. Generate PDF.	PDF contains inspection results & items.	Happypath

Test Case 15: RIS

These test cases validate the Requisition and Issue Slip (RIS) functionality, ensuring RIS creation, PDF generation, inventory adjustments, and proper handling of cancellation or updates.

Test Case ID	Feature	Test Scenario	Pre-conditions	Test Step	Expected Result	Type
RIS-001	RIS	Create RIS & Generate PDF	1. Items available. 2. Requester exists.	1. POST /requisition-issue-slip. 2. Generate PDF.	RIS created; inventory decremented; PDF valid.	Happypath
RIS-002	RIS	Cancel/Update RIS	1. RIS exists; not yet issued.	1. POST/DELETE to cancel RIS.	200; RIS cancelled; inventory adjustments reversed.	Happypath

Test Case 16: Notifications

These test cases validate the Notifications system, ensuring that status changes trigger notifications and emails correctly and that users can mark notifications as read or unread.

Test Case ID	Feature	Test Scenario	Pre-conditions	Test Step	Expected Result	Type
NOTIF-001	Notifications	Notification Status Change	1. PR/PO exists. 2. Mail configured.	1. Change status on PR/PO. 2. Assert Mail & Notification.	Mail queued; notification record created.	Happy path
NOTIF-002	Notifications	Notification Read/Unread	1. Notification exists.	1. GET notifications list. 2. POST read status.	200; notification read/unread toggled correctly.	Happy path

Test Case 17: PDF

These test cases validate PDF generation for PRs and POs, ensuring individual and bulk PDFs are correctly generated with accurate content and structure.

Test Case ID	Feature	Test Scenario	Pre-conditions	Test Step	Expected Result	Type
PDF-001	PDF	PR/PO PDF Generation	1. PR/PO exist.	1. Generate or preview PDF. 2. Validate content.	PDF includes correct fields & items.	Happypath
PDF-002	PDF	Bulk PDF Generation	1. Multiple PRs/POs exist.	1. Trigger bulk PDF generation. 2. Validate each PDF.	All generated PDFs are valid.	Happypath

DISCUSSION

LESSONS LEARNED

The development of the Web-based Supply and Property Management System presented several challenges that eventually drove tremendous technical and process maturity. One of the most critical lessons came from migrating legacy roles to the Spatie permission system-to avoid disrupting existing users, the team learned to place backward compatibility first with safety wrappers and precise migration scripts such as `roles:sync`. This defensive mindset was carried over into infrastructure, where fallbacks for activity logging and PDF generation were created to ensure that system stability was maintained, even when server-side dependencies or hosting environments proved brittle. On the user experience side, the team learned that reliability is equally important as functionality. To fight data loss and UI clutter, scattered status views were consolidated into a single management dashboard, and local storage persistence was implemented to prevent users from losing progress in long forms. This was complemented by a "defense-in-depth" validation strategy that married user-friendly frontend formatting with strict backend checks to ensure guaranteed data integrity. Finally, the project underscored the importance of continuous maintenance, which standardized coding patterns across modules and cleaned up obsolete seeders for tremendous streamlining of debugging and onboarding processes. Despite these successes, the team had to work within ongoing operational pressures, which meant

prioritizing tasks carefully-primarily core functionalities with plans for improvements sometime in future iterations.

FUTURE ENHANCEMENTS

Future Enhancement of the Supply and Property Management System can be improved on various fronts that might enhance efficiency, accuracy, or usability. The system can be upgraded to integrate RFID technology to automate the tracking of inventories without having to manually scan everything. It could facilitate the transfer of property between employees, making the transfer of assets within the institution smoother and with accurate records of custodianship. A notification system should enable users to receive alerts in real time for critical actions, such as low stock levels, pending approvals, or transfers that have been completed. Improved inventory reports should give decision-makers a better view of supplies used and their trends through advanced filtering and visualization, as well as options to export them. All this would make the system robust, user-friendly, and more supportive of the operational needs of the Supply and Property Management Office.

CONCLUSION

The development of the Supply and Property Management System for the Supply and Property Management Office of Camarines Norte State College successfully met its initial objectives by automating procurement, inventory tracking, and reporting processes. The system is a centralized, user-friendly platform that increases efficiency, accuracy, and accountability in the management of supplies and properties. The system replaced manual workflows from filing requisitions to approval hierarchies down to generating purchase orders with digitized workflows, addressing challenges in manual processing to reduce the risk of errors and delays. With additional

feature enhancements, such as role-based access, audit trails, and automatic report generation, the system ensures compliance with government regulations and significantly improves the decision-making process. Overall, the project solved the client's problem practically and in a sustainable manner, thereby helping the SPMO achieve streamlined operations, better resource management, and more transparent day-to-day operations.

The system is well set to accommodate further improvements and scalability. Potential future enhancements in the form of mobile access, QR code integration, advanced analytics, and cloud hosting would further make the system efficient, accessible, and responsive. This system thus has the potential to meet the operational needs of the SPMO not only in its present state but also in the future with respect to long-term sustainability and continued support for effective supply and property management within the college, through continued updates and user training.

REFERENCES

- College of Public Administration and Disaster Management, Nueva Ecija University of Science and Technology, Cabanatuan, Philippines, & Manuel, M. B. (2025). *Bridging procurement gaps in higher education: Addressing challenges, policy reforms, and digital transformation*. *International Journal of Advanced and Applied Sciences*, 12(7), 149–159. <https://doi.org/10.21833/ijaas.2025.07.015>
- Government Procurement Policy Board. (2024). *Republic Act No. 9184 – Implementing Rules and Regulations (IRR)*. Official Gazette.
- Gumilao, J. D. (2024). *Automated inventory management system for Department of Education Regional Office IX*. <https://doi.org/10.13140/RG.2.2.20268.48006>
- Lau, A. K. W. (2007). Educational supply chain management: A case study. *On the Horizon*, 15(1), 15–27. <https://doi.org/10.1108/10748120710735239>
- Madu, I. C., & Izuchukwu, M. C. (2022). Inventory management practices applied by stores officers in public tertiary educational institutions in southeast Nigeria. *Discovery*, 58(316), 370–377.
- Mncwango, B., & Ramdass, K. (2023, July 18). *Evaluating the efficiency of a procurement process at an institution of higher learning*. Proceedings of the 6th European International Conference on Industrial Engineering and Operations Management, Lisbon, Portugal. <https://doi.org/10.46254/EU6.20230381>
- Nasution, H. R. (2022). *Web-based information system for office equipment and inventory procurement at Bank Mandiri Jakarta Pusat*, 1(7).
- PhilGEPS. (n.d.). *Government's tool for procurement reforms*. <https://notices.philgeps.gov.ph>

Planergy. (2025). *Procurement problems in education: Challenges and solutions*.

<https://planergy.com/blog/problems-in-education-procurement/>

Ramos-Miller, M., & Pacheco, A. (2023). Towards inventory control excellence: An innovative approach based on a web-based platform. *F1000Research*, 12, 1471.

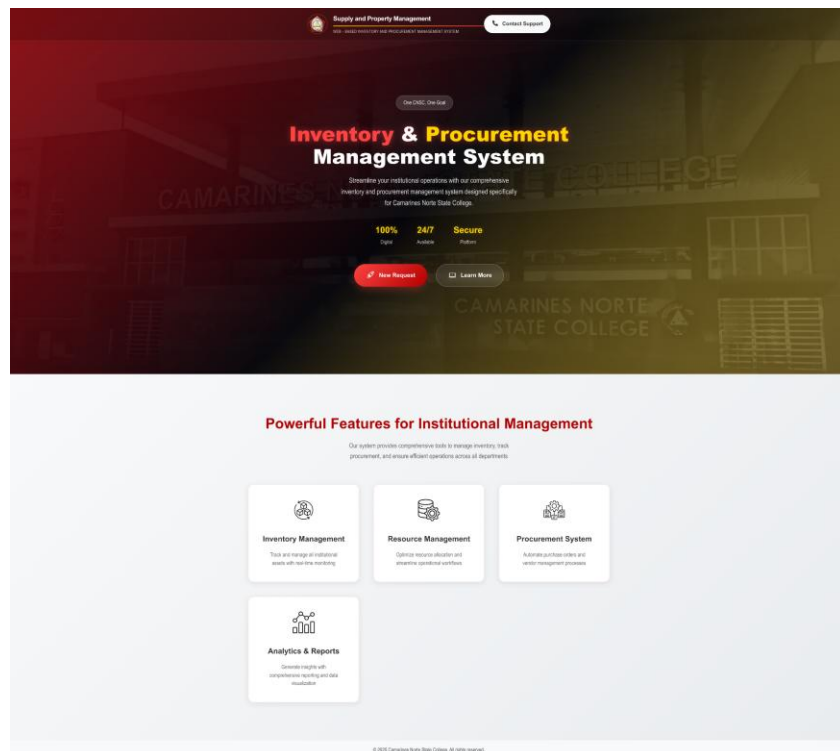
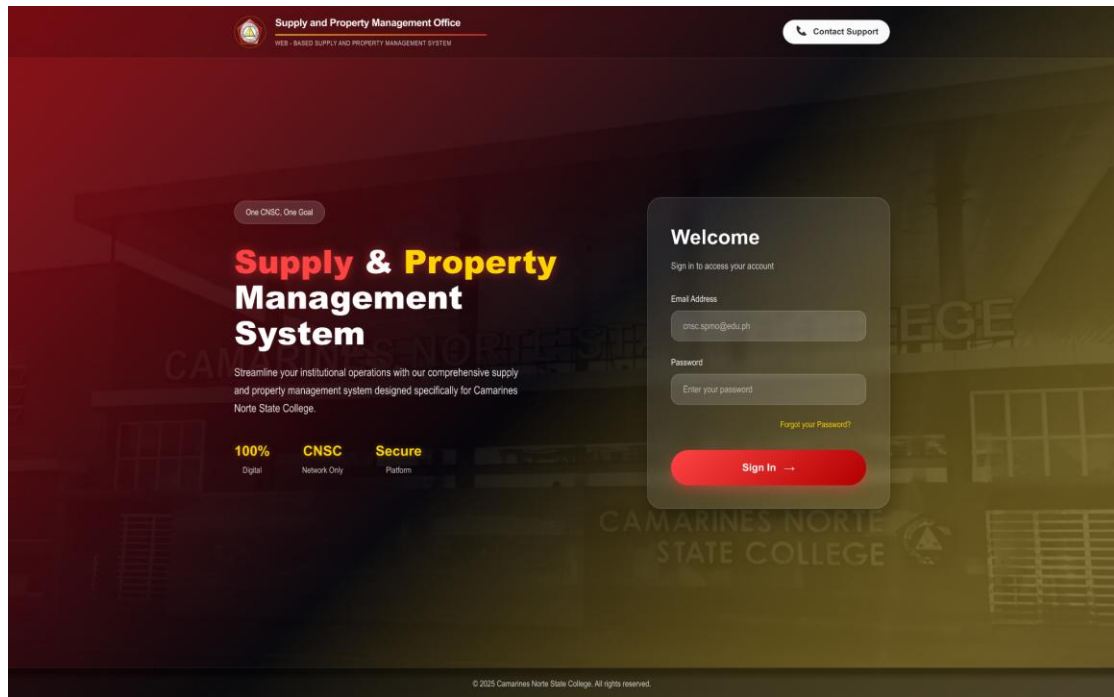
<https://doi.org/10.12688/f1000research.140745.1>

Young, R. R., Ruamsook, K., & Purdum, S. B. (n.d.). *Supply chain management series*.

Zhang, X., Sun, P., Xu, J., Wang, X., Yu, J., & Zhao, Z. (Year). *Blockchain-based safety management system for the grain supply chain*. IEEE.

APPENDICES. A

User Interface Design



Supply and Property Management Office
WEB-BASED SUPPLY AND PROPERTY MANAGEMENT SYSTEM

admin@example.com | Logout

PURCHASE REQUEST

1 Personal Info 2 Details & Priority 3 Review & Submit

Purchase Request of Office and Sanitorial Supplies

Submit an item acquisition for approval

Requester: VINCE STEVEN D. BALCE | Email Address: balcevince@gmail.com

Department: College of Education (CoEd) - Abalos Campus (Dart) | Designation: COLLEGE DEAN

Entity Name (optional): Camarines Norte State College

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PURCHASE REQUEST

1 Personal Info 2 Details & Priority 3 Review & Submit

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Submit an item acquisition for approval

Requester Name: VINCE STEVEN D. BALCE | Email Address: balcevince@gmail.com

Requester Position: COLLEGE DEAN

Department: College of Education (CoEd) - Abalos Campus (Dart)

Entity Name (optional): Camarines Norte State College

Item Details:

Item Description	Quantity	Unit	Estimated Price	Total Price
Office Supplies	100	Box	100.00	100.00
Sanitorial Supplies	50	Box	200.00	100.00

Requester's Signature: [Signature] | Requester's ID: [ID]

Approval Status: Pending

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WEB-BASED SUPPLY AND PROPERTY MANAGEMENT SYSTEM

admin@example.com | Logout

PURCHASE REQUEST

1 Personal Info 2 Details & Priority 3 Review & Submit

Purchase Request of Office and Sanitorial Supplies

Review Your Request

Requester Name: VINCE STEVEN D. BALCE | Email Address: balcevince@gmail.com

Requester Position: COLLEGE DEAN

Department: College of Education (CoEd) - Abalos Campus (Dart)

Entity Name (optional): Camarines Norte State College

Item Details:

Item Description	Quantity	Unit	Estimated Price	Total Price
Office Supplies	100	Box	100.00	100.00
Sanitorial Supplies	50	Box	200.00	100.00

Requester's Signature: [Signature] | Requester's ID: [ID]

Approval Status: Pending

Home | Back | Next | Cancel | Submit

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Dashboard Overview

Good morning, Administrator!

Welcome to your Supply System dashboard. Here's what's happening today.

Sunday, December 14, 2025
09:51 AM

Items

26

0% No change

Low Stock

11

+100% Up

Incoming

0

0% No change

Received Today

0

0% No change

Users

0

0% No change

Inventory Value

P253,575.00

+3% Up

Items by Category

Expendable, 46.15%

Non-Expendable, 22.00%

Semi-Expendable (High), 30.77%

Expendable, 22.00%

Non-Expendable, 22.00%

Semi-Expendable (High), 30.77%

Stock Status

In Stock

Low Stock

Out of Stock

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User Activities

Monitor user authentication and access history in real-time

TOTAL LOGIN RECORDS

3

All recorded logins

SUCCESSFUL LOGINS

3

Authenticated access

FAILED ATTEMPTS

0

Security alerts

UNIQUE USERS

1

Active accounts

Authentication History

Comprehensive log of all user login activities

Today

Total: 3

TIMESTAMP	USER	EMAIL ADDRESS	ACTION	DEVICE TYPE	IP ADDRESS	STATUS
18:11:21 2025-12-13	Administrator	admin@example.com	Login	Mozilla/5.0 (Windows NT 10.0; Win64; x64; AppleWebKit/537.36; Chrome/143.0.0.0; like Gecko) Chrome/143.0.0.0 Safari/537.36 Edg/143.0.0.0	192.168.1.1	SUCCESS
18:18:35 2025-12-13	Administrator	admin@example.com	Logout	Windows PC	192.168.1.1	SUCCESS
13:46:08 2025-12-13	Administrator	admin@example.com	Login	Mozilla/5.0 (Windows NT 10.0; Win64; x64; AppleWebKit/537.36; Chrome/143.0.0.0; like Gecko) Chrome/143.0.0.0 Safari/537.36 Edg/143.0.0.0	192.168.1.1	SUCCESS

Showing 1 to 3 of 3 entries

Previous1Next

User Actions

Log of actions performed by users (e.g., Create, Update, Delete)

TIMESTAMP	USER	ACTION	DETAILS
11:37:52 PM 12/10/2025	System	Stock Out: Binder Clips	Inventory ID: 2025-001 Transaction Ref: null SKU: B011 Quantity: 2 Product Name: Binder Clips

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List of Items

Manage item inventory

Expendable

Semi-Expendable

Non-Expendable

Search a Item

Sort By

Filter By

Rows: 10

ITEM ID	ITEM NAME	DESCRIPTION	QUANTITY	UNIT	UNIT COST	TOTAL VALUE	DATE	ACTION
E006	Alcohol Swabs	Sterile alcohol swabs, 100 pieces per box	60	boxes	P75.00	P4,500.00	2025-12-13	
E011	Binder Clips	Assorted size binder clips, 50 pieces per box	30	boxes	P45.00	P1,350.00	2025-12-13	
E001	Coffee	Ground coffee, 500g bag	20	bags	P180.00	P3,600.00	2025-12-13	
E010	Envelopes	White business envelopes, A4 size	100	pieces	P8.00	P800.00	2025-12-13	
E008	Highlighter Pens	Assorted color highlighter pens, 4-pack	40	packs	P55.00	P2,200.00	2025-12-13	
E007	Notebook	Spiral notebook, A5 size, 100 pages	60	pieces	P45.00	P2,700.00	2025-12-13	
E004	Paper Cups	Disposable paper cups, 200ml, 50 pieces per pack	40	packs	P55.00	P2,200.00	2025-12-13	
E005	Plastic Plates	Disposable plastic plates, 25cm, 50 pieces per pack	25	packs	P150.00	P3,750.00	2025-12-13	
E009	Sticky Notes	Yellow sticky notes, 100 sheets per pad	80	pads	P35.00	P2,800.00	2025-12-13	
E003	Sugar	White sugar, 1kg bag	15	bags	P45.00	P675.00	2025-12-13	

Showing 1 to 10 of 12 entries

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Add Item

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- Stock In
- Stock Out

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Stock In

Record incoming inventory and stock receipts

Add Stock In

Search stock transactions...

mm/dd/yyyy

Select Supplier

Sort By

Rows: 10

TRANSACTION ID	DATE	ITEM NAME	SKU	QUANTITY	UNIT COST	TOTAL COST	SUPPLIER	RECEIVED BY	ACTION
No records found									

No entries to display

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Stock Out

Record outgoing inventory and issued items

Issue Stock

Search stock issues...

Select Department

mm/dd/yyyy

X Clear

Export

Rows: 10

ISSUE ID	DATE	ITEM NAME	SKU	QUANTITY	UNIT COST	TOTAL COST	DEPARTMENT	ISSUED TO	ISSUED BY	ACTIONS
No records found										

No entries to display

Previous1Next

Stock Alert - Low Inventory

15 ITEMS

SKU	ITEM NAME	CATEGORY	CURRENT STOCK	UNIT	STATUS
N005	Conference Table	-	2	pieces	CRITICAL
SE006	Document Scanner	-	3	pieces	CRITICAL
SE003	Fire Extinguisher	-	5	pieces	CRITICAL
N002	Office Desk	-	5	pieces	CRITICAL
N004	Bookshelf	-	6	pieces	CRITICAL
SE005	Emergency Exit Sign	-	6	pieces	CRITICAL
SE008	Wireless Router	-	8	pieces	CRITICAL
N003	Filing Cabinet	-	8	pieces	CRITICAL
SE004	First Aid Sign	-	10	pieces	CRITICAL
N001	Office Chair	-	10	pieces	CRITICAL

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Supplier Management

Add, edit and manage suppliers

Add Supplier

NAME	ADDRESS	TIN	CONTACT	EMAIL	ACTION
No suppliers found					

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- Pending Approval
- Completed Request
- Status Management

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Procurement Panel

Create and manage new delivery requests

Search requests...

Filter by Status

All Status

Filter by Department

Select Department

Rows: 10

REQUEST ID	P.O. NUMBER	SUPPLIER	REQUEST DATE	DELIVERY DATE	TOTAL AMOUNT	STATUS	REQUESTED BY	DEPARTMENT	ACTION
No requests found									
Showing 0 entries									

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Pending Approval

Review and approve purchase requests

Search requests...

Filter by Status

All Status

Filter by Priority

All Priority

REQUEST ID	P.O. NUMBER	SUPPLIER	DELIVERY DATE	TOTAL AMOUNT	PRIORITY	STATUS	REQUESTED BY	DEPARTMENT	SUBMITTED DATE	ACTION
No pending requests found										

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Completed Request

View completed and archived purchase requests

Search requests...

Filter by Status

All Status

REQUEST ID	P.O. NUMBER	SUPPLIER	TOTAL AMOUNT	STATUS	REQUESTED BY	APPROVED BY	DELIVERED DATE	ACTION
No completed requests found								
TOTAL REQUESTS			TOTAL VALUE		COMPLETED REQUESTS			
0			P0.00		0			

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Status Management

Track and manage request statuses across all departments

PENDING
New requests received

RECEIVED
Awaiting processing

FINISHED
Successfully completed

CANCELLED
Request withdrawn

REJECTED
Request denied

RETURNED
Items sent back

Request Status Overview

Comprehensive list of all request statuses

Search by Requester ID, or ID

Filter by Department

Filter by Priority

REQUEST ID	REQUESTER	DESIGNATION	DEPARTMENT	ITEM	QUANTITY	UNIT	PRIORITY	DATE UPDATED	DETAILS	COST	REMARKS
No records											

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User Management

Monitor and manage system users and permissions

TOTAL USERS
Registered accounts

ACTIVE USERS
Currently active

INACTIVE USERS
Needs attention

System Users

Manage user accounts and permissions

NAME	EMAIL	ROLE	STATUS	CREATED
Administrator 1	admin@example.com	System Admin	ACTIVE	2025-12-13
Test User 2	test@example.com	Administrator	ACTIVE	2025-12-13

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Roles & Management

Manage team members, roles, and permissions

Team Members

MEMBER ID	NAME	EMAIL	ROLE	STATUS	CREATED	ACTIONS
1	Administrator Please login	admin@example.com	System Admin	ACTIVE	2025-12-13	
2	Test User Please login	test@example.com	Administrator	ACTIVE	2025-12-13	

System Roles

ROLE NAME	PERMISSIONS	ACTIONS
System Admin	MANAGE EVERYTHING MANAGE CATEGORIES MANAGE ITEMS MANAGE SUPPLIES MANAGE STOCK IN MANAGE STOCK OUT CREATE REQUESTS MANAGE REQUESTS VIEW REPORTS	
Administrator	MANAGE CATEGORIES MANAGE ITEMS MANAGE SUPPLIES MANAGE REQUESTS VIEW REPORTS	
Supply Officer	MANAGE ITEMS MANAGE SUPPLIES CREATE REQUESTS MANAGE REQUESTS VIEW REPORTS	
Supply Coordinator	MANAGE SUPPLIES MANAGE STOCK IN MANAGE STOCK OUT CREATE REQUESTS VIEW REPORTS	
Office Assistant	CREATE REQUESTS VIEW REPORTS	
Student Assistant	MANAGE STOCK IN MANAGE STOCK OUT CREATE REQUESTS	

**CMSC Supply
MANAGEMENT**

About Us
Learn more about the SPMO System and the team behind it

[Edit About Us](#)
Updated: 2023

- Dashboard
- Transactions
- Inventory Management
- Supplier Management

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- Acquisition**
 - Procurement Panel
 - Pending Approval
 - Completed Requests
 - Status Management

- Analytics
 - Reports
- Administration
 - User Management
- About Us**
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SS

SPMO System
Revolutionizing Supply & Property Management for Camarines Norte State College

Our Mission

To provide a comprehensive, user-friendly platform that streamlines supply management, property processes, and ensures transparency in resource allocation across all departments of CMSC.

Our Vision

To be the leading digital solution for educational institutions, setting the standard for efficient resource management, data-driven decision making, and operational excellence.

What We Offer
Comprehensive solutions for modern inventory management

Inventory Management
Real-time tracking of stock levels, automated alerts, and comprehensive inventory reports

Procurement Page
Streamlined purchase order creation, approval workflows, and vendor management

Analytics & Reporting
Powerful dashboards and customizable reports for data-driven decisions

Multi-User Access
Role-based permissions ensuring secure and organized collaboration

APPENDICES B.

GOVERNMENT FORMS

[illegible]

	A	B	C	D	E	F
1	Appendix 62					
2						
3	INSPECTION AND ACCEPTANCE REPORT					
4						
5						
6	Entity Name : _____			Fund Cluster : _____		
7						
8	Supplier : _____			IAR No. : _____		
9	PO No./Date : _____			Date : _____		
10	Requisitioning Office/Dept. : _____			Invoice No. : _____		
11	Responsibility Center Code : _____			Date : _____		
12						
13	Stock/ Property	Description	Unit	Quantity		
14						
15						
16						
17						
18						
19						
20						
21						
22						
23						
24						
25						
26						
27	INSPECTION			ACCEPTANCE		
28	Date Inspected : 10/11/2025			Date Received : 10/11/2025		
29						
30	Inspected, verified and found in order as to quantity and specifications			Complete		
31	☐			☐		
32				Partial (pls. specify quantity)		
33				☐		
34						
35						
36	Inspection Officer/Inspection Committee			Supply and/or Property Custodian		
37						
38						
39						

	A	B	C	D	E	F
1						Appendix 61
2						
3	PURCHASE ORDER					
4						
5	Entity Name					
6						
7	Supplier :			P.O. No. :		
8	Address :			Date :		
9	TIN :			Mode of Procurement :		
10	Gentlemen:					
11	Please furnish this Office the following articles subject to the terms and conditions contained herein:					
12						
13	Place of Delivery :			Delivery Term :		
14	Date of Delivery :			Payment Term :		
15	Stock/ Property No.	Unit	Description	Quantity	Unit Cost	Amount
16						
17						
18						
19						
20						
21						
22						
23						
24						
25						
26						
27						
28						
29						
30						
31						
32	(Total Amount in Words)					
33						
34	In case of failure to make the full delivery within the time specified above, a penalty of one-tenth (1/10) of one percent for every day of delay shall be imposed on the undelivered item/s.					
35						
36	Conforme:			Very truly yours,		
37						
38						
39						
40	Signature over Printed Name of Supplier			Signature over Printed Name of Authorized Official		
41						
42	Date			Designation		
43						
44						
45	Fund Cluster :			ORS/BURS No. :		
46	Funds Available :			Date of the ORS/BURS:		
47				Amount : _____		
48	Signature over Printed Name of Chief Accountant/Head of Accounting Division/Unit					
49						
50						

#	A	B	C	D	E	F	G	H
1								Appendix 59
2								
3								
4	INVENTORYCUSTODIAN SLIP							
5								
6	Entity Name: _____							
7	Fund Cluster : _____						ICS No : _____	
9	Quantity	Unit	Amount		Description	Inventory Item No.	Estimated Useful Life	
10			Unit Cost	Total Cost				
11								
12								
13								
14								
15								
16								
17								
18								
19								
20								
21								
22								
23								
24								
25								
26								
27								
28								
29								
30								
31								
32								
33								
34								
35								
36								
37								
38								
39								
40								
41								
42								
43								
44	Received from:					Received by:		
45								
46								
47	Signature Over Printed Name					Signature Over Printed Name		
48								
49	Position/Office					Position/Office		
50								
51	Date					Date		

INSPECTION AND ACCEPTANCE REPORT

Entity Name : _____

Fund Cluster : _____

[illegible]